Script-Aware Retrieval for Tanglish Queries over a Bilingual Tamil–English Document Base

M.E. Project Review – Modules 1 and 2 (2 November 2026)

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Abstract* - Millions of Tamil speakers type their questions in Tanglish: Tamil words written in Latin letters and freely mixed with English ("scholarship ku last date enna?"). The documents that answer them in Tamil Nadu are bilingual and not parallel: Government Orders and scheme notices often exist only in Tamil, while circulars and FAQs often exist only in English. Retrieval-Augmented Generation (RAG) systems are, however, built and tested on English queries. This paper reports Modules 1 and 2 of an M.E. thesis that measures what each RAG retrieval component does when the query is Tanglish. Module 1 defines the problem, surveys related work, states a narrowed research gap and designs a script-aware retrieval pipeline. Module 2 implements it on the public nlpctx/tamil-english-retrieval dataset: 500 Tamil-script news documents (with their English translations as the second half of a bilingual base) and 2,154 information needs, each written as an English, Tamil-script, romanised and code-mixed query. Five retrievers (word BM25, character 3-gram BM25, multilingual-e5-small, and two Reciprocal Rank Fusion hybrids) and two normalisers (the ITRANS scheme and a Tanglish rule set written for this project) were evaluated on a 2-core CPU. Tamil-script queries reach MRR@10 0.923, but romanised Tamil queries collapse to 0.143 with dense retrieval and 0.178 at best. Transliterating them with the Tanglish rules lifts character 3-gram BM25 to MRR@10 0.684 and Recall@10 0.812 ($p < 0.001$ under query and document-cluster bootstrap) for about 0.3 ms per query, and beats ITRANS on every retriever. Equal-weight BM25 + dense fusion, a common RAG default, hurts cross-script retrieval (English queries: 0.788 $\rightarrow$ 0.710). An audit shows that 22% of the dataset's "code-mixed" queries contain no Tamil word, which motivates the human-written bilingual Tanglish benchmark planned for Module 3.

***Keywords* —** *Retrieval-Augmented Generation (RAG), Code-Mixing, Tanglish, Tamil, Bilingual Retrieval, Transliteration, BM25, Dense Retrieval, Multilingual Embeddings, Reciprocal Rank Fusion.*

1. INTRODUCTION

A Retrieval-Augmented Generation (RAG) system answers a question in two steps [1]. A *retriever* first searches a document collection and returns the few documents most likely to hold the answer; a *generator*, a large language model (LLM), then writes the answer from them. If the retriever returns the wrong documents, the generator has nothing correct to work with, so retrieval quality sets the ceiling for the whole system.

Retrievers come in two families. *Sparse* retrievers such as BM25 [2] score documents by the query words they share, weighted by rarity. *Dense* retrievers encode queries and documents as vectors; multilingual models such as multilingual-E5 [3], BGE-M3 [4] and LaBSE [5] are trained so that text with the same meaning lands close together across languages. Many RAG systems combine both with Reciprocal Rank Fusion (RRF) [6].

Tamil has tens of millions of speakers, and in chat, search and messaging many of them do not type in Tamil script. They type romanised Tamil, often mixed with English: Tanglish. The same information need can therefore arrive in four forms (Table 1). The documents a Tamil Nadu help desk must search are bilingual: a Government Order (G.O.) may be published only in Tamil, a university circular only in English. Romanised Tamil shares neither the script of the Tamil documents nor the vocabulary of the English ones, and its spelling is unstandardised (*kurithu*, *kuritthu*, *kuriththu*).

Form	**Example query (from the dataset)**
English	Who is responsible for the Israel-Hamas conflict
Tamil script	இஸ்ரேல் ஹமாஸ் மோதலுக்கு யார் பொறுப்பு
Romanised Tamil	Israel Hamas mothalukku yaar poruppu
Code-mixed (Tanglish)	Israel Hamas conflict ku yaar responsible

TABLE 1 – ONE INFORMATION NEED IN FOUR QUERY FORMS

The common local fix is to translate the query into En-

glish and search English text; a Tamil Nadu scheme chatbot, SchemeAI [7], does exactly that over an English-only index. In a bilingual, non-parallel base this cannot reach an answer that exists only in Tamil. A general-purpose LLM such as ChatGPT reads Tanglish well in a chat, but a help desk must answer from thousands of documents, more than fit in a prompt, so the LLM still depends on a retrieval step. If that step collapses on romanised Tamil, the LLM never sees the right evidence.

This paper reports Modules 1 and 2 of the thesis. Its contributions are: (1) a problem definition and a factorial design for evaluating RAG retrieval components under four query forms over a bilingual Tamil–English document base; (2) a literature survey with a narrowed, explicitly stated research gap; (3) a CPU-only retrieval pipeline that reuses the retriever and metric design of the author's ChunkLab project [8]; (4) a Tanglish-to-Tamil-script rule transliterator compared with the ITRANS scheme; (5) real results for 2,154 queries $\times$ 8 query sets $\times$ 5 retrievers on the Tamil corpus and 4 query sets $\times$ 5 retrievers on the English corpus, with query-level and document-cluster bootstrap tests and latency; and (6) an audit showing that a public "code-mixed" Tamil query set is substantially English.

2. RELATED WORK

A. RAG, Retrievers and Rank Fusion

Lewis et al. [1] introduced RAG; BM25 [2] remains the lexical retriever in most RAG stacks. Query rewriting before retrieval [9] is the template for the LLM normaliser planned for Module 3. Cormack et al. [6] proposed RRF, which sums $1/(k + \text{rank})$ over ranked lists with $k = 60$. Later analyses are more cautious: Bruch et al. [10] found RRF parameter-sensitive and outperformed by a tuned convex combination, and Chauhan and Agarwal [11] showed that fusion ranks the relevant document below the better component on 13–48% of BEIR queries, a "fusion tax" that is largest when the dense retriever is clearly stronger. All of these analyses use queries and documents in the same script.

B. Multilingual Embeddings and Romanised Input

Multilingual embedders [3]–[5] are trained mainly on native-script text. Chari et al. [12] showed that multilingual dense retrievers, BGE-M3 included, deteriorate sharply on Latinised queries (Greeklish, Arabizi and similar), a "script gap" closed by fine-tuning on mixed native and Latinised queries; no Indic language was studied. Verma et al. [13] found that romanisation induces representations aligned with neither the native script nor English, a likely mechanism for the collapse reported in Section 5.

C. Mixed-Script and Code-Mixed Retrieval

Gupta et al. [14] formalised mixed-script IR for Hindi and raised MRR from 0.686 to 0.874 with cross-script query expansion. The FIRE Transliterated Search and Mixed Script IR tracks (2013–2016) ran retrieval subtasks on Hindi and Bengali and included Tamil only in word-level language labelling [15], [16], [17]. Rule-based normalisation [18] and transliteration plus character n-gram indexing [17] were already used there, so we do not claim either idea as new. The FIRE Code-Mixed IR track [19], [20] uses Bengali–English social-media posts, where fusion helps because queries and documents share the Roman script. Zeng et al. [21] showed drops of up to 27% for Chinese and Japanese code-switched queries.

D. Transliteration and Code-Mixed QA

Aksharantar and IndicXlit [22] provide open neural romanised-to-native transliteration, and IndicLID [23] identifies romanised Indic text; both report accuracy, not retrieval effect. Chandu et al. [24] crowd-sourced 2,848 Tamil–English ("Tamlish") factoid questions without a retrieval collection. For Hindi KBQA, back-transliteration beat translation by 5% and 35% [25]. Tamil code-mixed work has focused on sentiment and offensive-language classification [26], [27]. Recent code-mixed RAG studies cover Hinglish [28], [29]; Indi-RomCoM [30] and TanglishMedBench [31] evaluate LLMs on romanised Tamil without retrieval. Agarwal [32] found a hand-built normaliser redundant after noisy-query fine-tuning in a Hinglish NLU pipeline.

E. Tamil Retrieval, Bilingual Bases and Language Preference

Tamil retrieval benchmarks [33], [34] and the Sinhala–Tamil e-government study of Balasubramaniyam and Madushanka [35] use native-script queries only. Thenmozhi and Aravindan [36] built ontology-based Tamil–English CLIR by translating the query into English, the same family as SchemeAI [7]. Ranaldi et al. [37] compare translating the query with translating the evidence in multilingual RAG. Park and Lee [38] show that multilingual RAG prefers high-resource and query-language documents. The two closest papers to our bilingual framing are Amiraz et al. [39], who show cross-lingual retrieval loss and a balanced per-language fix for non-parallel Arabic–English corpora, and Hira et al. [40], who show English anchoring for romanised Hindi, Bengali, Marathi and Urdu code-switched queries over machine-translated parallel documents. Singh [41] and Chhoun et al. [42] provide the factorial and chunking methodology for English and Khmer RAG. Table 2 summarises the closest work.

F. Research Gap

What the surveyed work never combines is (a) romanised and code-mixed Tamil–English queries, eventually written by people; (b) a bilingual, non-parallel Tamil + English document base in which the answer may exist in only one language; (c) retrieval for RAG with modern dense, lexical and fusion components; and (d) a controlled comparison of those components and of query-normalisation front-ends on the same information needs. Language bias in mixed-

Work	Languages / script	Task and key result	Gap relative to this study
FIRE MSIR 2013–16 [15], [17], [16]; Gupta et al. [14]	Hindi, Bengali (Tamil only in labelling)	Mixed-script IR; MRR 0.686 → 0.874 (Hindi lyrics)	No Tamil retrieval; pre-dense, no RAG
FIRE CMIR [19], [20]	Bengali–English, Roman script	Code-mixed IR; best MAP 0.70; fusion helps	Same script for queries and documents
Zeng et al. [21]	Chinese / Japanese + English	Code-switching IR; drops up to 27%	Not Tamil; not romanised
Chari et al. [12]	Latinised zh, ru, ar, el, fa	Script gap in dense IR; closed by mixed training	No Indic language
Chauhan and Agarwal [11]; Bruch et al. [10]	English (BEIR)	Fusion hurts 13–48% of queries	No script mismatch
Chandu et al. [24]	Hinglish, Tamlish, Tenglish	2,848 Tamlish factoid questions	QA, no retrieval collection
Balasubramaniyam and Madushanka [35]	Sinhala, Tamil → English	BGE-M3 Recall@15 95.6% (Tamil)	Native-script queries only
Sharma et al. [29]; Gaikwad [28]	Hinglish	Code-mixed RAG; fine-tuned embedder 96.3%	Hindi only
SchemeAI [7]	English index, Tamil output	TN scheme chatbot (MiniLM + FAISS)	No Tamil documents, no Tanglish, no metrics
Amiraz et al. [39]	Arabic–English, non-parallel	Cross-lingual cost; balanced retrieval fix	Native script only; no Tamil
Hira et al. [40]	Romanised hi, bn, mr, ur; parallel MT docs	English anchoring in code-switched RAG	Parallel documents; no Tamil; dense only
nlpctx dataset [43]	Tamil, 4 query forms	LLM-generated retrieval data	No paper or baselines

TABLE 2 – CLOSEST PRIOR WORK AND THE GAP RELATIVE TO THIS STUDY

language indices has already been shown for Arabic–English [39] and romanised Hindi, Bengali, Marathi and Urdu [40], so our claim is narrower: Tamil, native non-parallel documents where a language preference becomes a hard miss, and model-free lexical remedies evaluated on CPU.

3. METHODOLOGY

A. Existing System

A typical Indian institutional RAG deployment splits documents into fixed-size chunks, embeds them with an English or multilingual model, optionally adds BM25 fused by RRF, and passes the query *as typed*. SchemeAI [7], a Tamil Nadu scheme chatbot, embeds questions with the English all-MiniLM-L6-v2 model against a FAISS index of one English CSV and translates only the output into Tamil. Its drawbacks are those of the existing approach: no script handling (a romanised query shares no characters with Tamil documents), fusion assumed safe, script-agnostic chunking, no Tamil documents, and no retrieval evaluation. Table 3 contrasts it with the proposed system.

B. Proposed System Architecture

The proposed system (Fig. 1) adds a script-aware front end and a measurement harness to a standard RAG retriever. The query path runs along the top row: token-level language identification, the normaliser, three retrievers and RRF fusion into a top-10 list. The corpus path builds the indexes from the bilingual document base. In Module 2 the chunker passes whole documents through; the Tamil-aware chunker (Module 3), the script-gated router and the LLM generator (Module 4) are dashed.

Aspect	Existing RAG / SchemeAI	Proposed system
Query forms	Typed text, one language	English, Tamil, romanised, code-mixed, each measured
Documents	English-only index	Bilingual Tamil + English base
Script handling	None (translate output only)	Token language ID + transliteration
Lexical	Word BM25 or none	Word + char 3-gram BM25
Dense model	English MiniLM	Multilingual mE5 (BGE-M3, LaBSE in M3)
Fusion	RRF by default	Measured per form; zero-score filter; router (M4)
Evaluation	None reported	2,154 queries × 4 forms, CIs, latency
Hardware	CPU	CPU only (2 cores)

TABLE 3 – EXISTING (SCHEMEAI-STYLE) VERSUS PROPOSED SYSTEM

C. Modules of the Thesis

Module 1 (complete): problem, literature survey, gap, existing vs proposed design and architecture. **Module 2 (complete):** dataset subset, query audit, normaliser, five retrievers, metrics, results and examples. **Module 3:** bilingual Tanglish benchmark of about 500 human-written questions with gold-language labels, mixed-index bias test, Tamil-aware chunking and stronger baselines. **Module 4:** script-gated router, LLM generation and answer evaluation. **Module 5:** paper, public benchmark and thesis.

D. Query Normalisation

Each query is split into runs of Latin letters and other characters. A Latin token is kept as English if its Zipf frequency in English (wordfreq) is at least 3.0 and it is not in a list of about 60 Tanglish particles (*ku, la, enna, pathi, panni,*

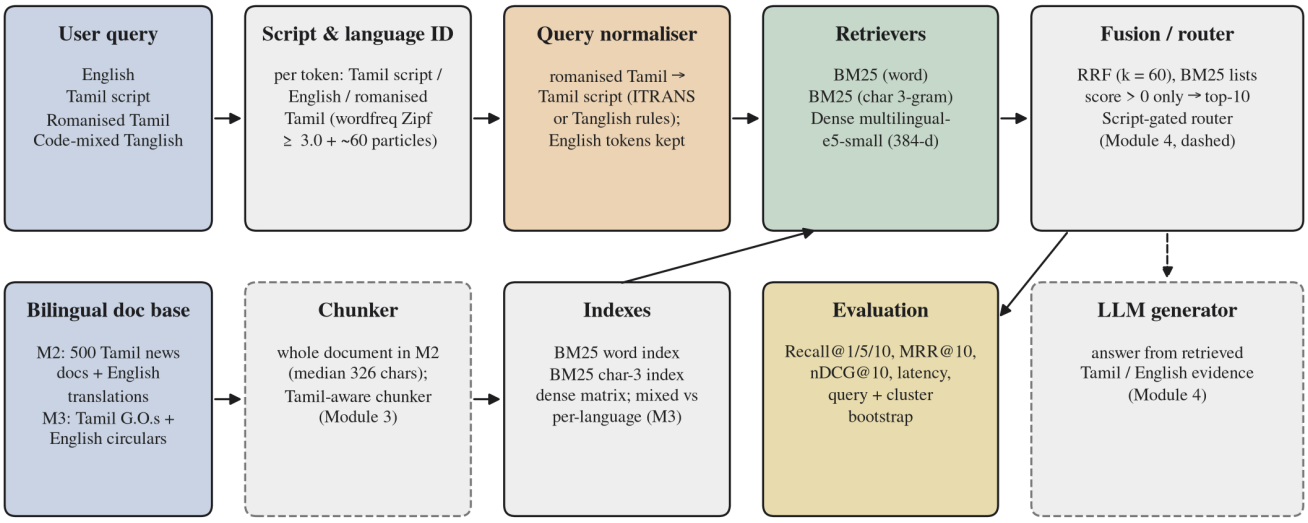


Figure 1: Proposed script-aware RAG retrieval architecture over a bilingual Tamil–English document base (solid = built in Modules 1–2; dashed = Modules 3–4)

yaar, …); otherwise it is treated as romanised Tamil and transliterated. Digits and punctuation pass through.

1. ITRANS mode applies the ITRANS-to-Tamil mapping of the indic-transliteration library. ITRANS is a formal Sanskrit-style scheme; for example it maps *e* to the long vowel ஏ and *n* to the dental ந everywhere, which does not match how Tanglish is typed.

2. Tanglish rules mode is a greedy longest-match transliterator written for this project from Tamil orthography alone, before any evaluation, and never tuned on test queries. Table 4 lists representative rules with the code's actual outputs. The rules cannot recover distinctions that Tanglish spelling does not encode (ல/ள typed as *l*; ன/ண/ந as *n*; த/ட as *t*), which is why character *n*-gram matching is also evaluated.

Tanglish	Tamil	Example
zh	ழ	kizhakku → கிழக்கு
th, dh	த	thadai → தடை
t, d	ட	tamil → டமில்
tr, rr	ற்ற	payitrapatta → பயிற்றபட்ட
ndr	ன்ற	endru → என்று
nth, ndh	ந்த	vandhu → வந்து
n (word start)	ந	neelam → நீலம்
other n; nn	ன; ன்ன	enna → என்ன
k, g, c, q	க	gangai → கங்கை
doubled consonant	C + pulli + C	kudikkalama → குடிக்கலம
aa, ee, oo, ai	vowel signs	maagaana → மாகான
final consonant	C + pulli	yaar → யார்

TABLE 4 – REPRESENTATIVE TANGLISH-TO-TAMIL RULES (LONGEST MATCH FIRST; ACTUAL CODE OUTPUT)

E. Retrievers

1. BM25 (word): Okapi BM25 (rank-bm25, $k_1 = 1.5$, $b = 0.75$) over lower-cased runs of Latin letters, digits and Tamil-block characters (U+0B80–U+0BFF). **2. BM25 (char 3-gram):** the same over boundary-marked character 3-grams (#word#), tolerant to spelling variation. **3. Dense:** multilingual-e5-small (118M parameters, 384-d) with "query:" and "passage:" prefixes, L2-normalised, ranked by dot product. **4. Hybrid-RRF:** RRF ($k = 60$) of the top-100 word-BM25 and dense lists, as in ChunkLab. **5. Hybrid-RRF3:** RRF of word BM25, char 3-gram BM25 and dense. Documents with a BM25 score of exactly 0 are removed before fusion.

Correction to an earlier run. The first version passed the full top-100 BM25 list to RRF. When a query matched nothing, the "ranking" was documents 0–99 in index order, which RRF rewarded; this affected 72% of English and 77% of code-mixed queries on the Tamil corpus. The corrected numbers are reported and the old ones are kept, labelled *naive*, for comparison.

F. Evaluation Metrics

Each query has one relevant document with rank $r(q)$ in the top 10.

- **Recall@k** (k = 1, 5, 10): fraction of queries whose relevant document is in the top k.
- **MRR@10**: mean of $1/r(q)$, 0 when absent.
- **nDCG@10**: mean of $1/\log_2(r(q)+1)$, 0 when absent.
- **Latency**: per-stage wall-clock time, one query at a time on 200 random queries; median and 95th percentile.

95% confidence intervals use 1,000 bootstrap resamples of queries. Paired differences use a two-sided paired bootstrap with 2,000 resamples. Because queries cluster within documents (about 4.3 per document), every key comparison is

repeated with a cluster bootstrap that resamples whole documents with all their queries.

4. EXPERIMENTAL SETUP

A. Dataset

The source is nlpctx/tamil-english-retrieval [43] (Apache-2.0): 9,299 LLM-generated query rows over 2,183 Tamil news documents, with IndicTrans2 English translations. To keep the study CPU-feasible, 500 documents were sampled with a fixed seed (42) and every query row for them kept, giving 2,154 queries, each with exactly one relevant document and all four forms, so comparisons are paired. Two corpora were built from the same 500 documents: the **Tamil corpus** (main setting) and the **English corpus** (the translations; secondary setting). Together they form the two halves of a bilingual base, although in this dataset they are parallel; Module 3 replaces them with a non-parallel collection. Documents were not chunked (median 326 characters). Table 5 gives the statistics.

Property	Value
Source	nlpctx/tamil-english-retrieval (Apache-2.0)
Full dataset	2,183 docs, 9,299 query rows, news
Documents sampled	500 (seed 42)
Queries per form	2,154
Query sets	4 raw + 4 normalised = 8
Relevant docs / query	1 (source document)
Doc length (chars)	mean 391, median 326, max 2,795
Corpora	Tamil script (main); English (secondary)
Ranked lists	84 runs × 2,154 = 180,936

TABLE 5 – EVALUATION SUBSET

B. Query Audit

Before scoring, the queries themselves were examined (Table 6). The romanised queries are genuinely Tamil (4.3 romanised-Tamil tokens per query; only 14% of their Latin tokens look English, mostly names). The "code-mixed" queries are mostly English: 1.74 Tamil tokens per query, 22.0% with none, 13.3% identical to the English query. Code-mixed results therefore mostly measure English-to-Tamil cross-lingual retrieval. After rule normalisation, 29.9% of romanised-query Tamil tokens occur in the corpus vocabulary, against 15.2% for ITRANS and 77.7% for native Tamil queries.

C. Protocol and Environment

Eight query sets were built (four raw; romanised and code-mixed each with ITRANS and rules). All five retrievers were run for the eight sets on the Tamil corpus (40 runs) and the four raw sets on the English corpus (20 runs); the two naive fusion variants were re-run on all 12 sets, giving 84 runs and 180,936 ranked lists. Hardware: 2 virtual ARM64 CPU cores, 7 GB RAM, no GPU. Software: Python 3.12, PyTorch (CPU), sentence-transformers, rank-bm25 0.2.2, indic-transliteration, wordfreq, pandas, NumPy and matplotlib.

Query set / measure	Tokens	Latin %	Tamil in vocab %
English	5.6	98.9	–
Tamil script	5.0	0.3	77.7
Romanised	5.1	98.9	0.0
Code-mixed	5.7	98.9	–
Rom → ITRANS	5.1	14.0	15.2
Rom → rules	5.1	13.8	29.9
CM → ITRANS	5.7	68.2	16.6
CM → rules	5.7	68.2	24.9
CM identical to EN		13.3%	
CM with no Tamil token		22.0%	
Tamil tokens / CM query		1.74	
Tamil tokens / rom. query		4.30	

TABLE 6 – QUERY AUDIT: TOKEN COMPOSITION (2,154 QUERIES PER SET)

5. RESULTS AND DISCUSSION

A. Raw Queries on the Tamil Corpus

Table 7 and Fig. 2 give the main results. With Tamil-script queries every retriever works (MRR@10 0.811–0.923). English queries are retrieved well only by the dense model (0.788), because BM25 cannot match English words to Tamil text; code-mixed queries behave like English (0.687). **Romanised queries fail everywhere**: dense MRR@10 is 0.143 and Recall@10 0.294, so for 7 in 10 romanised questions the correct document is not in the top 10; the best raw configuration reaches only 0.178. Character 3-grams beat word BM25 even on native Tamil (0.923 vs 0.811) because Tamil is agglutinative (மோதல், மோதலுக்கு, மோதலில்) and no stemmer was used; shared 3-grams act as an approximate stemmer.

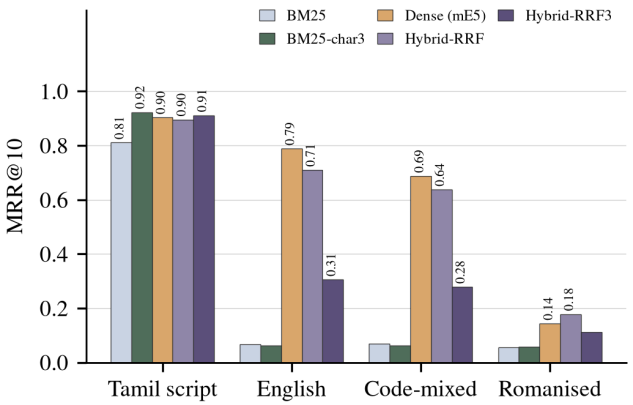


Figure 2: MRR@10 for each raw query form and retriever on the Tamil corpus

B. Effect of Query Normalisation

For romanised queries, normalisation is the largest effect in the study (Table 8, Fig. 3). Rules + char 3-gram BM25 raise MRR@10 from 0.058 to **0.684** and Recall@10 from 0.082 to **0.812**; rules + dense raise MRR@10 from 0.143 to

Query form	Retriever	R@1	R@5	R@10	MRR@10	MRR@10 95% CI	nDCG@10
Tamil script	BM25	0.764	0.872	0.891	0.811	[0.795, 0.825]	0.830
	BM25 char-3	0.898	0.954	0.964	**0.923**	[0.912, 0.933]	0.933
	Dense (mE5)	0.862	0.956	0.974	0.903	[0.893, 0.913]	0.921
	Hybrid-RRF	0.860	0.937	0.960	0.895	[0.884, 0.907]	0.911
	Hybrid-RRF3	0.883	0.945	0.970	0.911	[0.900, 0.921]	0.925
English	BM25	0.062	0.075	0.078	0.067	[0.057, 0.078]	0.070
	BM25 char-3	0.051	0.075	0.088	0.062	[0.053, 0.072]	0.068
	Dense (mE5)	0.711	0.892	0.937	**0.788**	[0.774, 0.803]	0.824
	Hybrid-RRF	0.567	0.889	0.936	0.710	[0.695, 0.726]	0.766
	Hybrid-RRF3	0.085	0.695	0.879	0.305	[0.295, 0.316]	0.442
Code-mixed	BM25	0.065	0.075	0.079	0.069	[0.059, 0.080]	0.072
	BM25 char-3	0.052	0.078	0.088	0.063	[0.054, 0.073]	0.069
	Dense (mE5)	0.597	0.805	0.862	**0.687**	[0.670, 0.706]	0.729
	Hybrid-RRF	0.505	0.806	0.863	0.637	[0.622, 0.655]	0.693
	Hybrid-RRF3	0.085	0.616	0.801	0.278	[0.268, 0.290]	0.403
Romanised	BM25	0.051	0.063	0.068	0.056	[0.046, 0.065]	0.059
	BM25 char-3	0.048	0.068	0.082	0.058	[0.048, 0.067]	0.064
	Dense (mE5)	0.093	0.212	0.294	0.143	[0.132, 0.157]	0.178
	Hybrid-RRF	0.129	0.247	0.321	**0.178**	[0.164, 0.192]	0.211
	Hybrid-RRF3	0.065	0.178	0.262	0.113	[0.103, 0.123]	0.148

TABLE 7 – RAW QUERIES ON THE TAMIL CORPUS (N = 2,154 PER FORM; BOLD = BEST MRR@10 PER FORM)

0.535; rules + RRF3 give the best Recall@10 of all (0.842). The Tanglish rules beat ITRANS on every retriever (char 3-gram: 0.684 vs 0.547). A gap to Tamil script remains (0.684 vs 0.923) because Tanglish spelling loses ல/ள, ன/ண/ந and த/ட. For code-mixed queries, normalisation helps the lexical retrievers (BM25 0.069 → 0.111; RRF3 0.278 → 0.420) but **does not change dense retrieval** (0.687 → 0.692, p = 0.45), since these queries contain little Tamil.

Query set	BM25	Char-3	Dense	RRF	RRF3	R@10 c3
Romanised	0.056	0.058	0.143	**0.178**	0.113	0.082
Rom. + ITRANS	0.210	**0.547**	0.328	0.386	0.506	0.705
Rom. + rules	0.333	**0.684**	0.535	0.537	0.627	0.812
Code-mixed	0.069	0.063	**0.687**	0.637	0.278	0.088
CM + ITRANS	0.093	0.197	**0.666**	0.585	0.412	0.322
CM + rules	0.111	0.238	**0.692**	0.568	0.420	0.369

TABLE 8 – EFFECT OF NORMALISATION: MRR@10 ON THE TAMIL CORPUS (LAST COLUMN: CHAR-3 RECALL@10)

C. Statistical Significance

Table 9 gives paired tests. Every improvement from the Tanglish rules for romanised queries, and the rules-over-ITRANS improvement, is significant ($p < 0.001$) under both the query and the stricter document-cluster bootstrap. The dense change for code-mixed queries is not ($p = 0.45$; cluster $p = 0.57$). The cluster bootstrap widens MRR@10 intervals by about 1.6 times (dense on English queries: [0.774, 0.803] becomes [0.766, 0.810]) but changes no conclusion.

D. Hybrid Fusion Can Hurt

Equal-weight hybrid retrieval is not robust across scripts (Table 10). For English or code-mixed queries against Tamil text, word BM25 matches nothing for most queries and only incidental tokens (*of, tamil, sri, from, news*) for the rest, yet RRF gives that list the same weight as the dense list. With zero-score documents excluded, RRF still lowers dense MRR@10 from 0.788 to 0.710 for English queries and from 0.687 to 0.637 for code-mixed queries (both $p < 0.001$); adding the char 3-gram list makes it far worse (0.305, 0.278). The naive run inflated the harm seven- to tenfold (0.788 → 0.187). Fusion helps when the lexical list carries signal: raw romanised queries rise from 0.143 to 0.178 because BM25 matches names and numbers. A code-mixed RAG system should therefore gate or weight retrievers by the query's script and never fuse zero-score lexical results.

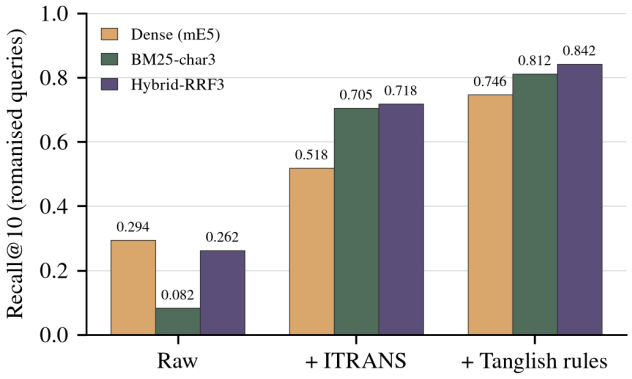


Figure 3: Recall@10 of romanised queries: raw, ITRANS-normalised and rule-normalised

E. English Half of the Bilingual Base

On the English-translation corpus (Table 11), English and code-mixed queries are easy (dense 0.939; RRF3 0.901 for code-mixed), Tamil-script queries are handled only by the dense model (0.742), and romanised queries are again

Comparison	Retriever(s)	MRR before	MRR after	Change	p (query)	p (cluster)	Change 95% CI (cluster)
Rom. → Rom.+rules	Dense	0.143	0.535	+0.391	< 0.001	< 0.001	[+0.365, +0.417]
Rom. → Rom.+rules	Char-3	0.058	0.684	+0.626	< 0.001	< 0.001	[+0.596, +0.655]
Rom. → Rom.+rules	RRF3	0.113	0.627	+0.514	< 0.001	< 0.001	[+0.485, +0.542]
Rom.+ITRANS → Rom.+rules	Dense	0.328	0.535	+0.207	< 0.001	< 0.001	[+0.184, +0.230]
Rom.+ITRANS → Rom.+rules	Char-3	0.547	0.684	+0.137	< 0.001	< 0.001	[+0.113, +0.161]
CM → CM+rules	Dense	0.687	0.692	+0.005	0.446	0.573	[-0.013, +0.022]
CM → CM+rules	BM25	0.069	0.111	+0.042	< 0.001	< 0.001	[+0.024, +0.062]
CM → CM+rules	RRF3	0.278	0.420	+0.142	< 0.001	< 0.001	[+0.115, +0.167]
Tamil: fusion	Dense → RRF	0.903	0.895	-0.008	0.168	0.287	[-0.023, +0.007]
Tamil: fusion	Dense → RRF naive	0.903	0.884	-0.019	0.003	0.032	[-0.036, -0.002]
English: fusion	Dense → RRF	0.788	0.710	-0.079	< 0.001	< 0.001	[-0.090, -0.068]
English: fusion	Dense → RRF naive	0.788	0.187	-0.601	< 0.001	< 0.001	[-0.634, -0.567]
CM: fusion	Dense → RRF	0.687	0.637	-0.050	< 0.001	< 0.001	[-0.062, -0.038]
CM: fusion	Dense → RRF naive	0.687	0.181	-0.506	< 0.001	< 0.001	[-0.540, -0.471]
Rom.: fusion	Dense → RRF	0.143	0.178	+0.034	< 0.001	< 0.001	[+0.023, +0.046]
Rom.: fusion	Dense → RRF naive	0.143	0.092	-0.052	< 0.001	< 0.001	[-0.075, -0.030]
Rom.+rules: fusion	Dense → RRF	0.535	0.537	+0.002	0.803	0.856	[-0.023, +0.026]
Rom.+rules: fusion	Dense → RRF naive	0.535	0.441	-0.094	< 0.001	< 0.001	[-0.126, -0.062]
Rom.+rules: fusion	Char-3 → RRF3	0.684	0.627	-0.057	< 0.001	< 0.001	[-0.080, -0.034]

TABLE 9 – PAIRED SIGNIFICANCE TESTS ON THE TAMIL CORPUS (2,000 RESAMPLES; CLUSTER = RESAMPLE THE 500 SOURCE DOCUMENTS)

Query form	Dense	RRF	RRF3	RRF naive	△ RRF
Tamil script	0.903	0.895	0.911	0.884	-0.008
English	0.788	0.710	0.305	0.187	-0.079
Code-mixed	0.687	0.637	0.278	0.181	-0.050
Romanised	0.143	0.178	0.113	0.092	+0.034
Rom. + rules	0.535	0.537	0.627	0.441	+0.002

TABLE 10 – FUSION VERSUS DENSE ALONE: MRR@10, TAMIL CORPUS (NAIVE = ZERO-SCORE BM25 DOCUMENTS FUSED)

the hardest form (best 0.557 with RRF3). Whatever the document language, romanised Tamil is the weakest query form, so a bilingual base does not remove the problem; it adds the question of which half a Tanglish query is drawn towards, which Module 3 tests.

Query form	BM25	Char-3	Dense	RRF	RRF3
Tamil script	0.064	0.050	**0.742**	0.728	0.113
English	0.870	0.876	**0.939**	0.920	0.916
Code-mixed	0.865	0.855	0.875	0.900	**0.901**
Romanised	0.522	0.445	0.333	0.544	**0.557**

TABLE 11 – ENGLISH-TRANSLATION CORPUS: MRR@10 (BOLD = BEST PER FORM)

F. Latency

Table 12 shows that the normaliser costs about 0.3 ms per query. Dense query encoding dominates (about 32 ms on 2 cores). The best romanised pipeline (rules + char 3-gram BM25) needs about 5 ms with no neural model, and the full rules + RRF3 pipeline about 38 ms, well within interactive limits.

Stage	Median ms	p95 ms
Tanglish rule normalisation	0.34	0.48
BM25 (word) + top-k	0.67	0.98
BM25 (char 3-gram) + top-k	4.88	8.59
Dense query encoding (CPU)	32.11	39.26
Dense search (500 × 384)	0.14	0.18
RRF fusion of 3 lists	0.19	0.23
Rules + RRF3 pipeline (sum of medians)	**38.3**	

TABLE 12 – PER-QUERY LATENCY ON 2 CPU CORES (200 RANDOM ROMANISED QUERIES)

G. Example Outputs

Table 13 shows real outputs of the dense retriever for three information needs. In Examples 1 and 2 the raw romanised query misses the top 10 and the rule-normalised query retrieves the gold document at rank 1. Example 3 shows the limits of rules: *Muthalaq* becomes முதலக் (not முத்தலாக்) and *masoda* becomes மசொட (not மசோதா), because the spelling marks neither the doubled *t* nor the long vowels, and dense retrieval still fails. A re-run for this review shows that char 3-gram BM25 on the same normalised query does rank the gold document first, which supports routing romanised queries to character matching. The code-mixed form also shows a language-ID failure: "Triple டலக் bill ரஜ்ய Sabha" splits one named entity across two scripts.

H. Discussion and Threats to Validity

A Tamil RAG system that passes Tanglish straight to a multilingual retriever misses most romanised questions; a few dozen rules costing under a millisecond recover most of the loss, and a character *n*-gram index absorbs residual spelling errors and Tamil suffixes. Hybrid fusion should be gated by script. The threats are: (1) every query is LLM-

	Form	Query given to the retriever	Rank	Top-1 retrieved document (Tamil, shortened)
Ex. 1	English	Colombia military soldiers killed in attack	1	✓ கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அர...
	Tamil script	தாக்குதலில் கொலம்பியா ராணுவ வீரர்கள் உயிரிழப்பு	1	✓ கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அர...
	Romanised	thakkudhalil Colombia raanuva thiraigal uyirizhappu	–	✗ பஞ்சாலை தொழிலாளர்களின் வாழ்வியலையும், உடுமலை வ...
	Rom. + rules	தக்குதலில் Colombia ரானுவ திரைகல் உயிரிழப்பு	1	✓ கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அர...
	Code-mixed	Colombia military soldiers killed in attack	1	✓ கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அர...
	CM + rules	Colombia military soldiers killed in attack	1	✓ கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அர...
Ex. 2	English	Northern Province school uniform length request	1	✓ வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீள...
	Tamil script	வடக்கு மாகாண பாடசாலை சீருடை நீளம் கோரிக்கை	1	✓ வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீள...
	Romanised	Vadakku maagaana paadasaalai seerudai neelam korikkai	–	✗ "பின்னே என்ன சார்? ஒருத்தரை வெறுக்கிறோம்னா அதை...
	Rom. + rules	வடக்கு மாகான பாடசாலை சீருடை நீலம் கொரிக்கை	1	✓ வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீள...
	Code-mixed	Northern province school uniform length increase request	1	✓ வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீள...
	CM + rules	Northern province school uniform length increase request	1	✓ வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீள...
Ex. 3	English	Triple Talaq Bill Rajya Sabha introduction	3	✗ மேற்கண்ட தகவல்கள் 2006 ஜூன் மல்லிகையில் மேமன்க...
	Tamil script	முத்தலாக் தடை மசோதா ராஜ்யசபா	1	✓ புதுடில்லி : 'முத்தலாக்' தடை மசோதா, ராஜ்யசபாவி...
	Romanised	Muthalaq thadai masoda Rajyasabha	–	✗ அழகிய மணிமுடியும், தாமரை மலர் போல சிவந்த திருக...
	Rom. + rules	முதலக் தடை மசொட ரஜ்யசப	–	✗ எந்தக் கட்சி ஆட்சியில் இருந்தாலும், 'நான் அடிப...
	Code-mixed	Triple Talaq bill Rajya Sabha la eppo	4	✗ எந்தக் கட்சி ஆட்சியில் இருந்தாலும், 'நான் அடிப...
	CM + rules	Triple டலக் bill ரஜ்ய Sabha ல எப்பொ	–	✗ எந்தக் கட்சி ஆட்சியில் இருந்தாலும், 'நான் அடிப...

TABLE 13 – EXAMPLE OUTPUTS OF THE DENSE RETRIEVER ON THE TAMIL CORPUS (RANK OF GOLD IN TOP 10; – = NOT RETRIEVED)

generated; (2) each query was generated from its own document, so lexical overlap is high and the 500-document corpus is small; (3) one relevant document per query makes scores a lower bound; (4) one small dense model was used; (5) the rules reflect one author; (6) the frequency-based language ID splits Indian named entities; (7) queries cluster within documents, handled by the cluster bootstrap; and (8) the Module 2 bilingual corpora are parallel translations, so language preference cannot yet be measured.

CONCLUSION AND FUTURE WORK

Modules 1 and 2 defined Tanglish retrieval for RAG over a bilingual Tamil–English base, surveyed related work and narrowed the gap, designed a script-aware pipeline, and evaluated it on 2,154 queries in four forms with five retrievers and two normalisers on a CPU. Romanised Tamil queries break standard retrieval (dense MRR@10 0.143 vs 0.923 for Tamil script); Tanglish rules with char 3-gram BM25 recover most of the loss (0.684, Recall@10 0.812) at about 0.3 ms per query and significantly beat ITRANS; equal-weight fusion is harmful across scripts (0.788 → 0.710); and the public code-mixed query set is English-heavy.

Module 3 (Dec 2026 – Jan 2027): about 500 human-written Tanglish questions with relevance judgements and gold-language labels over a bilingual, non-parallel collection of long Tamil G.O.s and scheme notices and English circulars and FAQs; a single mixed index vs per-language indices test of language preference [39], [40]; a Tamil-aware chunker; and baselines including IndicXlit [22], an LLM rewriter, a SchemeAI-style translate-to-English pipeline [7], BGE-M3, LaBSE and the Tanglish embedder Morgan-Tanglish-v7 [44]. **Module 4 (Feb – Mar 2027):** a script-gated router (dense for English-heavy queries, rules + char BM25 for romanised ones, fusion only when every list is informative), an LLM generator, and answer correctness and faithfulness with an LLM judge calibrated on human labels [45], [46]. **Module 5 (Apr – May 2027):** paper, public benchmark, code and thesis.

REFERENCES

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[2] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[3] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, and F. Wei, "Multilingual E5 text embeddings: A technical report," arXiv:2402.05672, 2024.

[4] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "BGE M3-Embedding: Multi-lingual, multi-functionality, multi-

granularity text embeddings through self-knowledge distillation," arXiv:2402.03216, 2024.

[5] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, "Language-agnostic BERT sentence embedding," in *Proc. 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, pp. 878–891.

[6] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009, pp. 758–759.

[7] M. Rekha, P. Pragadeeshwaran, S. Surendhar, and B. Udhayachandran, "Government schemes recommendation system API with bilanguage chatbot using RAG vector search," *Int. J. Advanced Research in Science, Communication and Technology (IJARSCT)*, vol. 6, no. 9, Apr. 2026, doi: 10.48175/IJARSCT-33228.

[8] S. PaulRaj K, "ChunkLab: The interactive RAG engineering playground," GitHub repository, 2026. [Online]. Available: https://github.com/selvin-paul-raj/chunklab

[9] X. Ma, Y. Gong, P. He, H. Zhao, and N. Duan, "Query rewriting in retrieval-augmented large language models," in *Proc. 2023 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, Singapore, 2023, doi: 10.18653/v1/2023.emnlp-main.322.

[10] S. Bruch, S. Gai, and A. Ingber, "An analysis of fusion functions for hybrid retrieval," *ACM Trans. Information Systems*, vol. 42, 2023, pp. 1–35, doi: 10.1145/3596512.

[11] A. Chauhan and A. Agarwal, "The fusion tax: A diagnostic analysis of when hybrid retrieval hurts," *IEEE Access*, vol. 14, 2026, pp. 159620–159637, doi: 10.1109/ACCESS.2026.3743502.

[12] A. Chari, I. Ounis, and S. MacAvaney, "Lost in transliteration: Bridging the script gap in neural IR," in *Proc. 48th Int. ACM SIGIR Conf. Research and Development in Information Retrieval (SIGIR '25)*, Padua, Italy, 2025, pp. 2900–2905, doi: 10.1145/3726302.3730226.

[13] A. A. K. Verma, A. Chatterjee, M. Gupta, and T. Chakraborty, "Multilingual language models encode script over linguistic structure," in *Proc. 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, San Diego, CA, USA, 2026, pp. 33685–33719, doi: 10.18653/v1/2026.acl-long.1554.

[14] P. Gupta, K. Bali, R. E. Banchs, M. Choudhury, and P. Rosso, "Query expansion for mixed-script information retrieval," in *Proc. 37th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2014, pp. 677–686, doi: 10.1145/2600428.2609622.

[15] R. Saha Roy, M. Choudhury, P. Majumder, and K. Agarwal, "Overview of the FIRE 2013 track on transliterated search," in *Proc. 5th Forum for Information Retrieval Evaluation (FIRE '13)*, ACM, 2013, pp. 1–7, doi: 10.1145/2701336.2701636.

[16] S. Banerjee, K. Chakma, S. K. Naskar, A. Das, P. Rosso, S. Bandyopadhyay, and M. Choudhury, "Overview of the mixed script information retrieval (MSIR) at FIRE-2016," in *Working Notes of FIRE 2016*, CEUR Workshop Proceedings, vol. 1737, 2016. [Online]. Available: https://ceur-ws.org/Vol-1737/T3-1.pdf

[17] R. Sequiera, M. Choudhury, P. Gupta, P. Rosso, S. Kumar, S. Banerjee, S. K. Naskar, S. Bandyopadhyay, G. Chittaranjan, A. Das, and K. Chakma, "Overview of FIRE-2015 shared task on mixed script information retrieval," in *Post Proceedings of the FIRE 2015 Workshops*, CEUR Workshop Proceedings, vol. 1587, 2015, pp. 19–25. [Online]. Available: https://ceur-ws.org/Vol-1587/T2-1.pdf

[18] D. Ganguly, S. Pal, and G. J. F. Jones, "DCU@FIRE-2014: Fuzzy queries with rule-based normalization for mixed script information retrieval," in *Proc. 6th Forum for Information Retrieval Evaluation (FIRE '14)*, ACM, 2014, pp. 80–85, doi: 10.1145/2824864.2824888.

[19] S. Chanda and S. Pal, "Overview of the shared task on code-mixed information retrieval from social media data," in *FIRE 2024 Working Notes*, CEUR Workshop Proceedings, vol. 4054, 2024, pp. 124–128. [Online]. Available: https://ceur-ws.org/Vol-4054/T2-1.pdf

[20] S. Chanda, K. Tewari, and S. Pal, "Overview of the CMIR track at FIRE 2025: Code-mixed information retrieval from social media data," in *Proc. 17th Annual Meeting of the Forum for Information Retrieval Evaluation (FIRE '25)*, ACM, 2025, pp. 13–16, doi: 10.1145/3777867.3778255.

[21] Q. Zeng, Y. Lu, Z. Zhou, H. Qi, P. Yu, F. Zhao, H. Yanaka, W. Xuan, and N. Yokoya, "Code-switching information retrieval: Benchmarks, analysis, and the limits of current retrievers," in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, doi: 10.18653/v1/2026.findings-acl.636.

[22] Y. Madhani, S. Parthan, P. Bedekar, G. NC, R. Khapra, A. Kunchukuttan, P. Kumar, and M. Khapra, "Aksharantar: Open Indic-language transliteration datasets and models for the next billion users," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, Singapore, 2023, pp. 40–57, doi: 10.18653/v1/2023.findings-emnlp.4.

[23] Y. Madhani, M. M. Khapra, and A. Kunchukuttan, "Bhasha-Abhijnaanam: Native-script and romanized language identification for 22 Indic languages," in *Proc. 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, Toronto, Canada, 2023, doi: 10.18653/v1/2023.acl-short.71.

[24] K. Chandu, E. Loginova, V. Gupta, J. van Genabith, G. Neumann, M. Chinnakotla, E. Nyberg, and A. W. Black, "Code-mixed question answering challenge: Crowd-sourcing data and techniques," in *Proc. Third Workshop on Computational Approaches to Linguistic Code-Switching*, Melbourne, Australia, 2018, pp. 29–38, doi: 10.18653/v1/W18-3204.

[25] V. Gupta, M. Chinnakotla, and M. Shrivastava, "Transliteration better than translation? Answering code-mixed questions over a knowledge base," in *Proc. Third Workshop on Computational Approaches to Linguistic Code-Switching*, Melbourne, Australia, 2018, doi: 10.18653/v1/W18-3205.

[26] B. R. Chakravarthi, V. Muralidaran, R. Priyadharshini, and J. P. McCrae, "Corpus creation for sentiment analysis in code-mixed Tamil-English text," in *Proc. 1st Joint Workshop on Spoken Language Technologies for Under-resourced Languages (SLTU) and Collaboration and Computing for Under-Resourced Languages (CCURL)*, 2020, pp. 202–210.

[27] B. R. Chakravarthi, R. Priyadharshini, V. Muralidaran, S. Suryawanshi, N. Jose, E. Sherly, and J. P. McCrae, "Overview of the track on sentiment analysis for Dravidian languages in code-mixed text," in *Proc. 12th Annual Meeting of the Forum for Information Retrieval Evaluation (FIRE '20)*, 2020, pp. 21–24, doi: 10.1145/3441501.3441515.

[28] S. S. Gaikwad and J. S. Gaikwad, "Reducing hallucinations in retrieval-augmented LLM question answering for Indian code-mixed queries," in *Smart Trends in Computing and Communications (SmartCom 2026)*, Lecture Notes in Networks and Systems, vol. 1991, Cham: Springer, 2026, pp. 402–415, doi: 10.1007/978-3-032-27754-1_37.

[29] S. Sharma, R. Rathee, N. Mishra, and S. Bhatt, "Multilingual embedders for explainable Hinglish retrieval in RAG," in *Proc. Int. Conf. Artificial Intelligence and Cyber Security (ICAICS 2025)*, Lecture Notes in Networks and Systems, vol. 2024, Cham: Springer, 2026, pp. 157–172, doi: 10.1007/978-3-032-28379-5_13.

[30] A. Das, M. Parmar, M. Ramnath, and P. Verma, "IndiRomCoM: Code-mixed benchmark for evaluating LLMs on romanized Indic-English instructions," arXiv:2606.30790, 2026.

[31] S. S. Anthea, N. R. Mary, and K. Varalakshmi, "TanglishMedBench: A benchmark for evaluating LLMs on Tamil-English code-mixed medical queries," Zenodo dataset, v1.0.0, 2026, doi: 10.5281/zenodo.22308075.

[32] P. Agarwal, "Where does noise robustness live? Evidence from ablation, role swaps, and cross-lingual transfer in code-mixed banking NLU," in *Proc. 11th Workshop on Noisy and User-generated Text (W-NUT)*, 2026, pp. 356–367, doi: 10.18653/2026.wnut-1.26.

[33] S. Haq, A. Sharma, O. Khattab, N. Chhaya, and P. Bhattacharyya, "IndicIRSuite: Multilingual dataset and neural information models for Indian languages," in *Proc. 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, Bangkok, Thailand, 2024, doi: 10.18653/v1/2024.acl-short.46.

[34] P. Prasanjith, P. More, A. Kunchukuttan, and R. Dabre, "IndicRAGSuite: Large-scale datasets and a benchmark for Indian language RAG systems," arXiv:2506.01615, 2025.

[35] D. Balasubramaniyam and T. Madushanka, "Query translation vs. cross-lingual embeddings for Sinhala-Tamil e-government information retrieval," arXiv:2608.12820, 2026.

[36] D. Thenmozhi and C. Aravindan, "Ontology-based Tamil–English cross-lingual information retrieval system," *Sādhanā*, vol. 43, art. no. 157, 2018, doi: 10.1007/s12046-018-0942-7.

[37] L. Ranaldi, B. Haddow, and A. Birch, "Multilingual retrieval-augmented generation for knowledge-intensive question answering task," in *Findings of the Association for Computational Linguistics: EACL 2026*, Rabat, Morocco, 2026, pp. 697–716, doi: 10.18653/v1/2026.findings-eacl.35.

[38] J. Park and H. Lee, "Investigating language preference of multilingual RAG systems," in *Findings of the Association for Computational Linguistics: ACL 2025*, Vienna, Austria, 2025, pp. 5647–5675, doi: 10.18653/v1/2025.findings-acl.295.

[39] C. Amiraz, Y. Fyodorov, E. Haramaty, Z. Karnin, and L. Lewin-Eytan, "The cross-lingual cost: Retrieval biases in RAG over Arabic-English corpora," in *Proc. Third Arabic Natural Language Processing Conf. (ArabicNLP 2025)*, 2025, pp. 69–83, doi: 10.18653/v1/2025.arabicnlp-main.6.

[40] M. Hira, J. Mire, and M. Sap, "The language of evidence: Robustness and bias in code-switched RAG," in *Proc. 2026 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2026, pp. 41664–41685. [Online]. Available: https://aclanthology.org/2026.emnlp-main.1830

[41] S. K. Singh, "Parser, chunking, and embedding interactions in retrieval-augmented generation over Indian government regulatory documents," arXiv:2609.31660, 2026.

[42] S. Chhoun, P. Po, S. Ros, W.-S. Cho, and S. Khoeurn, "Evaluation of chunking strategies for effective text embedding in low-resource language on agricultural documents," arXiv:2605.22203, 2026.

[43] nlpctx, "Tamil-English code-mixed retrieval evaluation dataset (tamil-english-retrieval)," Hugging Face Datasets, 2026, Apache-2.0. [Online]. Available: https://huggingface.co/datasets/nlpctx/tamil-english-retrieval

[44] Vishnu N, "Morgan-Tanglish-v7: Sentence embeddings for romanised Tamil-English code-mixed text," Zenodo preprint, Jun. 2026, doi: 10.5281/zenodo.20792177.

[45] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "RAGAs: Automated evaluation of retrieval augmented generation," in *Proc. 18th Conf. European Chapter of the Association for Computational Linguistics: System Demonstrations*, St. Julians, Malta, 2024, doi: 10.18653/v1/2024.eacl-demo.16.

[46] R. Hada, V. Gumma, A. de Wynter, H. Diddee, M. Ahmed, M. Choudhury, K. Bali, and S. Sitaram, "Are large language model-based evaluators the solution to scaling up multilingual evaluation?," in *Findings of the Association for Computational Linguistics: EACL 2024*, St. Julian's, Malta, 2024, doi: 10.18653/v1/2024.findings-eacl.71.